

Route study

Christopher Coad, Director of Business Services, requested assistance to study the efficiency of the bus routes. School transportation is a critical part of Brecksville Broadview Heights's education services. To retain the current level of excellence, the following will be analyzed:

- Current bus schedules
- Student ridership for each building, current and future
- School buildings bell schedules, current and future
- Alternative routing options for the new elementary campus
- Estimated cost of each routing option

Methodology

In general, costs of operation in a school bus fleet are tied directly to the number of buses operated daily. Typically, the use of more buses results in higher cost, both for capitalization of the vehicle purchase and for total cost of operation. Efficiency studies completed with other school districts and with districts that have a large geographical area have found it more cost effective for one school bus to make two or more trips to school than it is for two buses to make one trip to school.

School buses have two limiting factors: seating capacity and route time. Route time is influenced not only by the time between building bell schedules but also by efficient use of group bus stops. Less time is required for stopping to load and unload at group or neighborhood stops. The time required to load or unload students at school must also be accounted for when planning adequate time for the buses to complete their routes. Bus parking, parent parking and the time needed for students to walk to the buses are the major components for management in schools' loading/unloading areas.

Route planning begins with the afternoon tiers due to fully utilizing bus capacity as each building dismisses students because all riders require a seat at the same time. The length of time required to transport students home and return for the next building to dismiss establishes the amount of route time for each tier. The starting time for each building is then adjusted to create the educational length of day.

Routing Logistics

Routing Scenario in FY2019

The district was using a modified quad-tier routing system with some buses performing two or three tiers. The total number of buses used was 36.

Other factors relevant to this study:

- Policy designates children living beyond the following walking limits shall be entitled to bus transportation:
 - Kindergarten at Noon one (1) mile
 - Kindergarten in Morning, or afternoon one (1) mile
 - Grades 1 through 8 one (1) mile
 - Grades 9 through 12 one (1) mile with one-half (1/2) mile walking distance cluster stops.
 - Exceptions to the foregoing limits may be made based on financial constraints of the District
 - Bus routes shall be established so that an authorized bus stop is available within reasonable walking distance of the home of every transported resident student.
 - District transports an estimated 220 students that reside within one (1) of their assigned building.
- The school district covers 29 square miles with rider density of 78 students per square mile, with 2,262 students transported by bus to school.

District realignment routing scenario

- Central School was closed in June of 2019. The fifth-grade students are attending the Middle school. The fourth-grade students are attending the elementary schools.
- A triple-tier routing are implemented at the beginning of FY 2020.
- The High school bell schedule has been moved back by 15 minutes in the morning and afternoon.
- The Middle school bell schedule has been moved back by 20 minutes in the morning and by 15 minutes in the afternoon.
- The elementary schools' bell schedule has been advanced by 15 minutes in the morning but has remained unchanged in the afternoon.

Goals identified for this study are:

- 1) Analyze the current routing to assess ridership and routing times.
- 2) Provide routing options for this current school year that increase efficiency and bus safety.

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- 3) Assess the cost for each routing option for the current school year.
- 4) Provide logistical options for the opening of the new elementary school.
- 5) Provide an estimated annual operating expense for each option.

Determining Ridership

In any routing study, the number of students that ride the buses must be identified first. We used district routing software to perform this task that identified: 83% of elementary students are bus riders, 85% of middle school students are bus riders and 40% of high school students are bus riders. BBHCSD staff has requested that due to the change in the latchkey program for grades 4 and 5, expect 90% of elementary students will be bus riders. Ridership numbers for High and Middle Schools students will be unchanged.

Optimal routing occurs when you can schedule similarly sized groups of students at intervals that allow buses to run multiple routes. In practice, there will always be a certain tier that has more students and requires more buses. Sometimes this occurs because of nonpublic, special education or vocational routing needs. Sometimes it is purely a result of the number of students that are grouped at a single time. As part of the iterative planning process, the planner then focuses on that peak tier and works to either reduce the ridership count or plan extra routing capacity for that tier. The two best ways to make those adjustments are changes in bell times or changes in the number of buses used.

The third way is to change bus ridership is to only transport students that reside in the eligibility zone. Ridership can also decrease by increasing the distance from the schools where the eligibility begins. Either option will reduce the 220 students that the district currently transports that reside less than a mile from their school.

Current routing plan

Plan A

BBHCSD is using 40 buses, as of August 23, 2019 in a modified three-tier scenario as follows:

Morning transportation

- 23 buses transport regular ed high school students and three buses transport special needs students to school.
- 30 buses transport regular ed middle school students to school and two buses transport special needs students to school.
- 29 buses transport regular ed elementary students to school, four transport special ed students and two buses transport pre-school students.
- Five buses transport non-public students.

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Afternoon transportation

- 26 buses transport high school students and two buses transport special needs students home.
- 30 buses transport middle school students and two buses transport special needs students home.
- 31 buses transport elementary students four transport special ed students and two buses transport pre-school students home.
- Eight buses transport non-public students home.

Routing options and tier arrangement

The following options are a best-case scenario regarding the number of buses used to transport students on each tier and the cost for that option. There may be factors beyond the control of the transportation department, such as; road closures, student loading management or administrative student assignments that will require the use of more buses.

Plan B – a one year plan

This plan modifies school bell schedules by assigning the elementary schools to start and dismiss first. HS follows and the middle school is the third tier. The route time in the afternoon benefits from the shorter elementary educational day. The route time for the elementary tier is 60 minutes. The option uses 34 buses in a three-tier scenario as follows:

Morning transportation – 34 buses are required.

- 30 buses transport the elementary school students.
- 21 buses transport high school students.
- 25 buses transport the middle school students.
- Six buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students to CVCC.

Afternoon transportation – 34 buses are required.

- 27 buses transport elementary school students home, four elementary buses are scheduled to omit a high school route.
- 21 buses transport high school students home.
- 25 buses transport middle school students home.
- Four buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students from CVCC.

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Plan C

In this plan, we examine the option to return to the two-tier scenario using 37 buses and retaining the starting order of the buildings. The routing plan is as follows:

Morning transportation

- 33 buses transport high and middle school students concurrently. To fully utilize bus capacity, an average of 48 students will be assigned to each bus.
- 29 buses transport elementary students to school.
- Five buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students to CVCC.

Afternoon transportation

- 29 buses transport high and middle school students home from school with each bus averaging 51 students.
- Three high/middle school buses will be scheduled to exclude the elementary tier to accommodate long first tier routes.
- 29 buses transport elementary students home from school.
- Four buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students from CVCC.

Plan D – this plan can be implemented for the new elementary campus also.

In this plan, we examine the two-tier scenario with scheduling elementary students on the first tier both in the morning and afternoon. This plan requires 39 buses. The routing plan is as follows:

Morning transportation – 45-minute route time.

- 34 buses transport elementary students (83% ridership) to school with an average of 38 riders assigned to each bus.
- 31 buses transport high and middle school students to school. Bus capacity is nearly fully utilized with an average of 47 students assigned to each bus.
- Six buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students to CVCC.

Afternoon transportation – 60-minute route time.

- 31 buses transport elementary students (83% ridership) home with an average of 43 riders assigned to each bus.
- 33 buses transport high and middle school students home with each bus averaging 44 students.

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- Five buses transport non-public students.
- Four buses transport students with special needs.
- Three high/middle school routes are scheduled to run longer routes and do not perform elementary routes.
- One bus transports students from CVCC.

Plan E – 3-tier option for the opening of the new elementary campus.

This plan schedules elementary school first in the morning and afternoon. The three-tier option provides bus capacity that allows flexibility for an increase in ridership without adding additional buses. The route scenario is:

Morning transportation – requires 36 buses

- 32 buses transport elementary school students to school with an average of 37 riders assigned to each bus.
- 23 buses transport high school students to school with an average of 25 riders assigned to each bus.
- 25 buses transport middle students to school with an average of 34 riders assigned to each bus.
- Six buses transport non-public students, some nonpublic schools will be shuttled but three will be on dedicated buses.
- Four buses transport students with special needs.
- One bus transports students to CVCC.

Afternoon transportation – requires 36 buses.

- High school dismissal bell times will be set 60 minutes later than the elementary.
- 28 buses transport elementary school students home, with 7 buses scheduled to exclude the high school tier, allowing for an extended elementary route.
- 21 buses transport high school students home with an average of 25 riders assigned to each bus.
- 28 buses transport middle students home with an average of 32 riders assigned to each bus.
- Four buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students from CVCC.

Plan F– 2-tier option for the opening of the new elementary campus.

In this plan, we examine the option to return to the two-tier scenario requiring 39 buses and start the high/middle schools first. The routing plan is as follows:

- Morning transportation

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- 35 buses transport high and middle school students together.
- 29 buses transport elementary students to school. Bus capacity will be utilized with an average of 45 students assigned to each bus.
- Six buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students to CVCC.

Afternoon transportation

- 31 buses transport high and middle school students home with each bus averaging 41 students.
- 32 buses transport elementary students home from school.
- Five buses transport non-public students.
- Four buses transport students with special needs.
- One bus transports students from CVCC.

Fleet charts

To illustrate the impact of the routing plans, we have included a fleet assignment chart for each plan that list buses and assignments. These charts are based upon the routing models completed to determine the validity of our assumptions on time and potential for the buses to complete their assignments.

The logistical options have been reviewed and the most viable plans are presented to the Board of Education for consideration. The remaining logistical options have been included in this report to be used for comparison.

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Cost analysis

The cost impact associated with each routing plan is prepared by using the operating cost of each bus as reported on the FY2019 ODE T2. For each bus that is taken out of service in the routing plan, there is a savings associated. This savings is less than the actual cost of the bus due to residual overhead costs.

The tier cost for each bus is \$2,240 if there is no change in the length of overall route service time in the morning or afternoon route set.

- 1) The current routing plan (Plan A) using 40 buses. The annual estimated cost of operation for this plan will be \$3,067,661.
- 2) Implementing Plan B, a three-tier routing plan using 34 buses, has an estimated operational cost of \$2,586,045 annually.
- 3) Implementing Plan C, a two-tier routing plan using 37 buses, has an estimated operational cost of \$2,843,309 annually.
- 4) Implementing Plan D, (elementary first tier) a two-tier routing plan using 39 buses, has an estimated operational cost of \$2,884,664 annually.
- 5) Implementing Plan E, (elementary first tier) a three-tier routing plan using 36 buses, has an estimated operational cost of \$2,792,423 annually.
- 6) Implementing Plan F, (high/middle schools first tier) a two-tier routing plan using 39 buses, has an estimated operational cost of \$2,980,928 annually.

The estimated cost impacts are only for transportation's operational costs, and do not reflect the capitalization cost of bus purchases.

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Recommendations for consideration

- 1) To reduce costs for one year, implement Plan B as soon as practical.
- 2) For the opening of the new elementary campus, implement a change in the bell schedules that arrange more route time in the afternoon and reduce morning layover time between routes.
- 3) Consider implementing Plan D, a two-tier option, that schedules elementary campus on the first tier.
- 4) Consider implementing Plan E, a three-tier option, that schedules elementary campus on the first tier, and provides flexibility in bus capacity for a possible increase in ridership.
- 5) Continue to use group bus stops for all grade levels to minimize the route time for each tier and minimize bus riding time for students.
- 6) Consider bus parking arrangements that reduce the walking distance for students and prioritize bus egress to increase route time efficiency and students' safety.
- 7) The district should consider parents of bus riders subscribing for bus service to plan efficient routes.
- 8) School bus drivers should be counting students monthly and monitor regular bus attendance thoroughly.
- 9) Students that are identified as inactive bus riders should be changed to non-rider status in the routing software for accurate route sheet information.
- 10) Only active riders should be scheduled for the bus routes.
- 11) The bell schedules of all buildings must be correspondingly adjusted. A change in one building bell schedule without a corresponding time for the remaining buildings will have an impact on the bus route time.

Thank you for the opportunity to provide support for the Brecksville Broadview Heights City School District. We welcome the opportunity to help with issues or tasks related to this report if further assistance is desired.